

```

    }
    });
    producer.flush();
    producer.close();
}
}

```

Java consumer: The Dashboard class implements the Kafka consumer application and it runs at the ABC Operations Centre. It listens to *abc-bus-location* with the group ID *abc-dashboard* and displays the location details from different buses as soon as messages are available. Here, too, many details are hard coded which are otherwise supposed to be configured:

```

public static void main(String[] args) {
    String broker = "127.0.0.1:9092";
    String groupId = "abc-dashboard";
    Properties props = new Properties();
    props.setProperty(ConsumerConfig.BOOTSTRAP_SERVERS_
CONFIG, broker);
    props.setProperty(ConsumerConfig.KEY_DESERIALIZER_
CLASS_CONFIG,
        StringDeserializer.class.getName());
    props.setProperty(ConsumerConfig.VALUE_DESERIALIZER_
CLASS_CONFIG,
        StringDeserializer.class.getName());
    props.setProperty(ConsumerConfig.GROUP_ID_CONFIG,
groupId);

    @SuppressWarnings("resource")
    Consumer<String, String> consumer = new
KafkaConsumer<String, String>(props);
    consumer.subscribe(Arrays.asList("abc-bus-location"));
    while (true) {
        ConsumerRecords<String, String> records
            = consumer.poll(Duration.ofMillis(1000));

        for (ConsumerRecord<String, String> record :
records) {
            String topic = record.topic();
            int partition =
record.partition();
            String key = record.
key();
            String value = record.
value();
            System.out.
println(String.format(
                "Topic=%s,
Partition=%d, Key=%s, Value=%s",
                topic, partition,
key, value));

```

```

    }
    }
}
}

```

Dependencies: A JDK of 8+ version is required to compile and run this code. The following Maven dependencies in the *pom.xml* download and add the required Kafka client libraries to the *classpath*:

```

<dependency>
    <groupId>org.apache.kafka</groupId>
    <artifactId>kafka-clients</artifactId>
    <version>2.8.0</version>
</dependency>
<dependency>
    <groupId>org.slf4j</groupId>
    <artifactId>slf4j-simple</artifactId>
    <version>1.7.25</version>
</dependency>

```

Deployment

As the topic *abc-bus-location* is created with three partitions, it makes sense to run three consumers to read the location updates quickly. For that, run the Dashboard in three different terminals simultaneously. Since all the three instances of Dashboard register with the same group ID, they form a group. Kafka attaches each Dashboard instance with a specific partition.

Once the Dashboard instances are up and running, start the *Fleet* on a different terminal. Figure 6, Figure 7, and Figure 8 are sample console messages on the Dashboard terminals.

A closer look at the console messages reveals that the consumers on the first, second and third terminals are reading messages from *partition-2*, *partition-1*, and *partition-0*, in that order. Also, it can be observed that the messages with the keys BLRHBL002, BLRHBL004 and BLRHBL006 are written into *partition-2*, the messages with the key BLRHBL005 are written into *partition-1*, and the remaining are written into *partition-0*.

```

Topic=abc-bus-location, Partition=2, Key=BLRHBL006, Value=13.024843, 77.546983
Topic=abc-bus-location, Partition=2, Key=BLRHBL004, Value=13.659576, 76.944675
Topic=abc-bus-location, Partition=2, Key=BLRHBL002, Value=14.399654, 76.045834
Topic=abc-bus-location, Partition=2, Key=BLRHBL006, Value=13.024843, 77.546983
Topic=abc-bus-location, Partition=2, Key=BLRHBL004, Value=13.659576, 76.944675
Topic=abc-bus-location, Partition=2, Key=BLRHBL002, Value=14.399654, 76.045834
Topic=abc-bus-location, Partition=2, Key=BLRHBL006, Value=13.024843, 77.546983
Topic=abc-bus-location, Partition=2, Key=BLRHBL004, Value=13.659576, 76.944675
Topic=abc-bus-location, Partition=2, Key=BLRHBL002, Value=14.399654, 76.045834
Topic=abc-bus-location, Partition=2, Key=BLRHBL006, Value=13.024843, 77.546983
Topic=abc-bus-location, Partition=2, Key=BLRHBL004, Value=13.659576, 76.944675
Topic=abc-bus-location, Partition=2, Key=BLRHBL002, Value=14.399654, 76.045834

```

Figure 6: Dashboard Terminal - 1